

The offset-four family of regular sequences of power sums in three variables

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Abstract

Let $p_m = x^m + y^m + z^m \in \mathbb{C}[x, y, z]$. We prove that, for every integer $r \geq 1$, the sequence (p_r, p_{r+1}, p_{3r+4}) is regular if and only if $r \not\equiv 1 \pmod{3}$. A uniform reduction identifies the relevant locus as $e_2^2 = 4e_1e_3$, where e_1, e_2, e_3 are the elementary symmetric polynomials. We classify its common projective zeros with p_r, p_{r+1} : they exist exactly when $r \equiv 1 \pmod{3}$ and are the projective permutations of $[1 : \omega : \omega^2]$, with ω a primitive cube root of unity. The proof uses a rational parametrization, a modulus comparison, and a Laurent identity followed by a uniform distance estimate.

1 Statement and context

Throughout, set

$$S = \mathbb{C}[x, y, z], \quad p_m = x^m + y^m + z^m \quad (m \geq 1),$$

and write

$$e_1 = x + y + z, \quad e_2 = xy + xz + yz, \quad e_3 = xyz.$$

We use the usual ordered definition of a regular sequence: its ideal is proper and each entry is a nonzerodivisor modulo the preceding entries.

Theorem 1. *For every integer $r \geq 1$, the sequence (p_r, p_{r+1}, p_{3r+4}) is regular in S if and only if $r \not\equiv 1 \pmod{3}$.*

The supplied preprint of Abedelfatah [1], Theorem 1.1 and Corollary 1.2, treats the families (p_r, p_{r+1}, p_{3r+j}) for $j \in \{0, 1, 2, 3\}$. Its Proposition 2.3 gives four reductions modulo (p_r, p_{r+1}) . One further recurrence step reduces the offset $j = 4$ to a quartic curve. The principal step here is the complete common-zero classification on that curve. The first modulus comparison is adapted from the $e_2 = 0$ argument in Proposition 3.1 of that preprint; the subsequent Laurent identity and distance estimate handle the quartic parametrization.

Power-sum generators also occur in Zeng's work on bosonic–fermionic coinvariant rings [2], which supplies adjacent invariant-theoretic and exact-computation context. Our comparison with prior results is limited to the supplied preprint and this adjacent reference. Broader literature priority has not been established, and we make no claim that the family was previously open. The theorem concerns exactly the offset-four family; it does not settle the general power-sum regular-sequence conjecture discussed in [1].

2 Reduction to a quartic curve

We recall two elementary facts, also stated in [1, Lemmas 2.1–2.2].

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Lemma 2. *Three positive-degree homogeneous forms in S form a regular sequence if and only if they have no common zero in $\mathbb{P}^2(\mathbb{C})$. Moreover, every common projective zero of p_r, p_{r+1} has $xyz \neq 0$.*

Proof. By the homogeneous Nullstellensatz, absence of a common projective zero means that the ideal of the three forms has radical (x, y, z) . The forms then constitute a homogeneous system of parameters in the three-dimensional Cohen–Macaulay ring S , hence a regular sequence in the displayed order. Conversely, a regular sequence of length three generates an ideal of height three, so its homogeneous zero set contains no projective point. Positive degree ensures that the ideal is proper.

For the second assertion, suppose $z = 0$. If $y \neq 0$, then $a = x/y$ satisfies $a^r = a^{r+1} = -1$. Division gives $a = 1$, a contradiction. If $y = 0$, then $x = 0$ as well. Permutation treats the other coordinates. $\square$

Lemma 3. *For every integer $r \geq 1$,*

$$p_{3r+4} \equiv e_3^r(4e_1e_3 - e_2^2) \pmod{(p_r, p_{r+1})}. \quad (1)$$

Proof. Work in $S/(p_r, p_{r+1})$ and put $u = x^r, v = y^r, w = z^r$. Then

$$u + v + w = 0, \quad xu + yv + zw = 0, \quad uvw = e_3^r.$$

For $j \geq 0$, define $Q_j = x^j u^3 + y^j v^3 + z^j w^3$. As in [1, Proposition 2.3], the first three values are

$$Q_0 = 3uvw, \quad Q_1 = e_1uvw, \quad Q_2 = e_2uvw. \quad (2)$$

Here the first equality is the sum-of-cubes identity. For the second, the relation $u + v + w = 0$ gives

$$u^2 - vw = v^2 - uw = w^2 - uv.$$

Multiplying $xu + yv + zw = 0$ by this common value gives $Q_1 - e_1uvw = 0$. For the third, expansion gives

$$0 = (xu + yv + zw)(xu^2 + yv^2 + zw^2) = Q_2 - e_2uvw;$$

for example, the mixed xy terms sum to $xyuv(u + v) = -xyuvw$.

Each of x, y, z satisfies $T^3 - e_1T^2 + e_2T - e_3 = 0$. Therefore

$$Q_{j+3} = e_1Q_{j+2} - e_2Q_{j+1} + e_3Q_j \quad (j \geq 0).$$

Using (2) gives

$$Q_3 = 3e_3uvw, \quad Q_4 = (4e_1e_3 - e_2^2)uvw.$$

Since $Q_4 = p_{3r+4}$ and $uvw = e_3^r$, this proves (1). $\square$

By Lemmas 2 and 3, at a common projective zero of p_r, p_{r+1} the equation $p_{3r+4} = 0$ is equivalent to

$$e_2^2 = 4e_1e_3. \quad (3)$$

3 Classification on the quartic

Proposition 4. *Fix $r \geq 1$ and a primitive cube root of unity ω . The common zero set of $p_r, p_{r+1}, e_2^2 - 4e_1e_3$ in $\mathbb{P}^2(\mathbb{C})$ is empty if $r \not\equiv 1 \pmod{3}$. If $r \equiv 1 \pmod{3}$, it consists exactly of the projective permutations of $[1 : \omega : \omega^2]$.*

Proof. Let $[x : y : z]$ be such a common zero. All three coordinates are nonzero by Lemma 2. Permute them so that $|x| \geq |y| \geq |z|$ and scale to $x = 1$. Choose a square root t of y . Substituting $y = t^2$ in (3) gives

$$\begin{aligned} 0 &= ((1 + t^2)z + t^2)^2 - 4(1 + t^2 + z)t^2z \\ &= ((t - 1)^2z - t^2)((t + 1)^2z - t^2). \end{aligned}$$

Changing the sign of t exchanges the factors, so we may write

$$(x, y, z) = \left(1, t^2, \frac{t^2}{(1 + t)^2}\right), \quad t \neq 0, -1. \quad (4)$$

This covers the possible degeneracies: if a coefficient $(t \pm 1)^2$ vanishes, that factor is the nonzero constant $-t^2$, so the other factor supplies the representation. The coordinate ordering gives

$$0 < \rho := |t| \leq 1, \quad |1 + t| \geq 1. \quad (5)$$

Subtracting zp_r from p_{r+1} in (4) and multiplying by $(1 + t)^2$ yields

$$2t + 1 + t^{2r+3}(t + 2) = 0. \quad (6)$$

Write $a = \operatorname{Re} t$. The second inequality in (5) implies $a \geq -\rho^2/2$. The modulus identity used in [1, Proposition 3.1, p. 6] now gives

$$\begin{aligned} |2t + 1|^2 - \rho^2|t + 2|^2 &= (1 - \rho^2)(1 + \rho^2 + 4a) \\ &\geq (1 - \rho^2)^2 \geq 0. \end{aligned} \quad (7)$$

Since $t + 2 \neq 0$ when $|t| \leq 1$, equation (6) implies

$$\rho^{2r+3} = \frac{|2t + 1|}{|t + 2|} \geq \rho.$$

Together with $0 < \rho \leq 1$ and $r \geq 1$, this forces $|t| = 1$.

Set $s = |1 + t|^2$. Then $1 \leq s \leq 4$ and

$$s = 2 + t + t^{-1}, \quad (1 + t)^2 = st.$$

Scaling (4) by $1/t$ gives the coordinates $(t^{-1}, t, 1/s)$. With $A_n = t^n + t^{-n}$, the original two equations become

$$A_r = -s^{-r}, \quad A_{r+1} = -s^{-r-1}. \quad (8)$$

Direct expansion proves the Laurent identity

$$A_r^2 + A_{r+1}^2 - (t + t^{-1})A_r A_{r+1} = 4 - (t + t^{-1})^2. \quad (9)$$

Indeed, the terms of degrees $\pm 2r$ and $\pm(2r + 2)$ cancel, leaving $2 - t^2 - t^{-2}$ on both sides. Substitution of (8) and $t + t^{-1} = s - 2$ into (9) gives

$$\frac{2s + 1}{s^{2r+2}} = s(4 - s),$$

or equivalently the necessary condition

$$s^{2r+3}(4 - s) = 2s + 1. \quad (10)$$

If $1 < s \leq 3$, then

$$\begin{aligned} s^{2r+3}(4 - s) - (2s + 1) &\geq s^3(4 - s) - (2s + 1) \\ &= (s - 1)(s^2(3 - s) + 3s + 1) > 0, \end{aligned}$$

contradicting (10). The value $s = 4$ would give $0 = 9$. It remains to rule out $3 < s < 4$ using the original equation (8). In this interval, equation (10) gives

$$|t - 1|^2 = 4 - s = \frac{2s + 1}{s^{2r+3}} < \frac{9}{3^{2r+3}} = \frac{1}{3^{2r+1}}.$$

The identity $t^r - 1 = (t - 1)(1 + t + \cdots + t^{r-1})$ and $|t| = 1$ imply

$$|t^r - 1|^2 \leq r^2 |t - 1|^2 < \frac{r^2}{3^{2r+1}} \leq \frac{1}{3}. \quad (11)$$

The last inequality follows from $r \leq 3^r$ for all $r \geq 1$: it holds at $r = 1$, and $r + 1 \leq 3r$ proves the inductive step. On the other hand, equation (8) gives

$$|t^r - 1|^2 = 2 - A_r = 2 + s^{-r} > 2,$$

a contradiction. Thus $s = 1$.

It follows that $t + t^{-1} = -1$, so $t^2 + t + 1 = 0$. The coordinates $(t^{-1}, t, 1/s)$ are therefore a permutation of $(1, \omega, \omega^2)$. Conversely, this triple satisfies $e_1 = e_2 = 0$ and hence (3), and

$$1 + \omega^m + \omega^{2m} = \begin{cases} 0, & 3 \nmid m, \\ 3, & 3 \mid m. \end{cases}$$

Consequently $p_r = p_{r+1} = 0$ there exactly when $r \equiv 1 \pmod{3}$. This proves both the classification and its stated parameter range. $\square$

Proof of Theorem 1. If $r \not\equiv 1 \pmod{3}$, Lemma 3 and the nonvanishing of e_3 in Lemma 2 would put every common projective zero of p_r, p_{r+1}, p_{3r+4} on (3). Proposition 4 excludes all such points, so the sequence is regular by Lemma 2. If $r \equiv 1 \pmod{3}$, none of $r, r + 1, 3r + 4$ is divisible by three. The point $[1 : \omega : \omega^2]$ is then a common zero of all three power sums, so the sequence is not regular. This includes $r = 1$. $\square$

4 Computational scope and assistance

The proof above is uniform in r and covers all complex projective points. It uses no finite computation. As supplementary checks, exact algebraic elimination was carried out for every $r \leq 60$ with $r \not\equiv 1 \pmod{3}$. This includes 32 resultant cases with $13 \leq r \leq 60$. A separate verifier reconstructed normalized Newton polynomials as traces of powers of a companion matrix, checked the stored rational Bézout identities for $1 \leq r \leq 12$, and certified nonvanishing of the later resultants by degree-preserving reductions over the prime fields of orders 1009, 1013, and 1019. These checks certify the stated finite nonvanishing; they do not verify every decimal digit of a stored integer resultant or establish the infinite theorem. The supplementary scripts and their exact outputs are retained with the research materials.

OpenAI Codex assisted with proof development, symbolic verification, a separate automated mathematical review, and manuscript preparation. The exact computations used Python and SymPy. Automated mathematical review is distinct from external peer review and from formal proof-assistant verification.

References

- [1] A. Abedelfatah, *Two Infinite Families of Regular Sequences of Power Sums in Three Variables*, preprint, arXiv:2609.07932v1 (2026). <https://arxiv.org/abs/2609.07932v1>.
- [2] Zijian Zeng, *THE $n = 5$ Case of Lentfer's $(1, 2)$ -Bosonic-Fermionic Coinvariant Basis Conjecture*, preprint (2026). <https://doi.org/10.2139/ssrn.7379218>.